

SS3 GEOGRAPHY LESSON NOTES

FIRST TERM

WEEK 1: DENUDATIONAL PROCESSES

1.1 MEANING OF DENUDATION

Denudation is a comprehensive term used in geomorphology to describe all the processes that wear away, strip bare, or lower the land surface of the Earth's crust. It is the sum total of processes that cause the disintegration and transportation of rock and soil. Denudation plays a crucial role in landscape development, continuously re-shaping the Earth's topography.

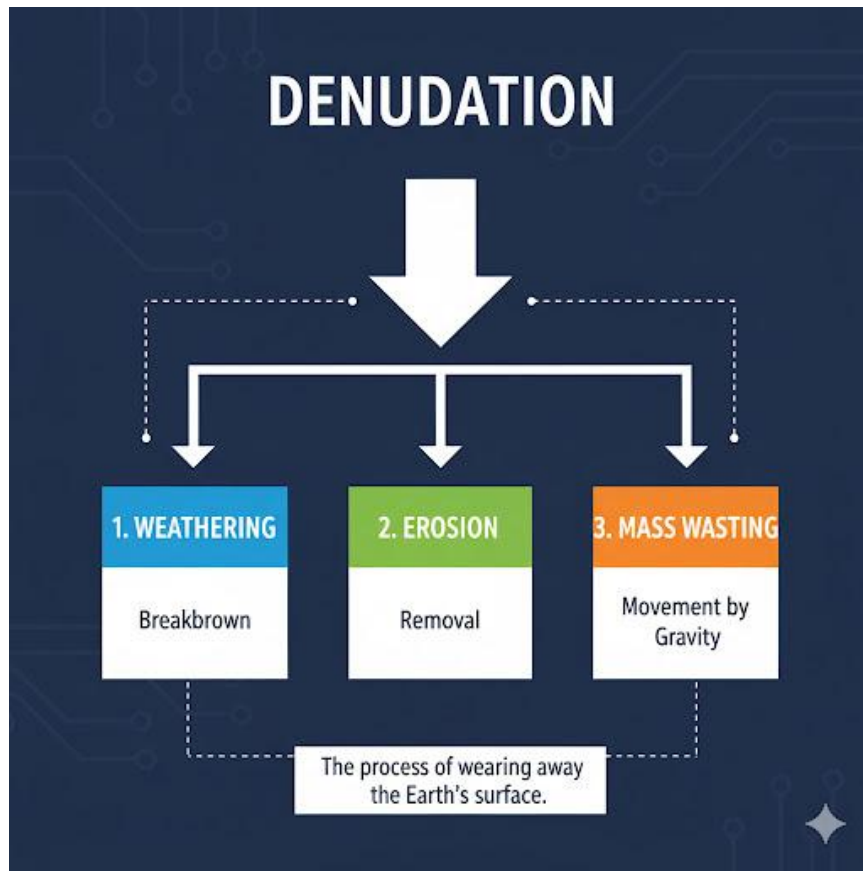
The term is derived from the Latin word *denudare*, meaning "to strip bare."

1.2 COMPONENTS OF DENUDATION

Denudation is not a single process, but an umbrella term that includes three main components operating in sequence:

1. **Weathering:** The *in-situ* (in place) disintegration (breaking apart) or decomposition (chemical decay) of rocks at or near the Earth's surface. It does not involve movement.
2. **Erosion:** The *removal* and *transportation* of weathered material (like soil, sand, and rock fragments) from one place to another. This process requires a mobile agent.
3. **Mass Wasting (Mass Movement):** The downslope movement of a large mass of rock or soil *primarily under the influence of gravity*. It is a form of transport, but the agent is gravity itself.

The full process involves **Transportation** (the movement of the eroded material) and **Deposition** (the laying down of the material when the agent loses energy).



1.3 DISTINCTION BETWEEN WEATHERING AND EROSION

This is a fundamental concept. Though both are part of denudation, they are different.

Aspect	Weathering	Erosion
Definition	The <i>in-situ</i> (in place) breakdown of rocks.	The <i>removal and transportation</i> of weathered materials.
Movement	No movement or transportation is involved.	Movement is essential. The material is carried away.
Agents	Primarily driven by climatic elements: temperature (sun), precipitation (water), air, and biological agents (plants, animals).	Driven by dynamic agents: running water (rivers), wind , moving ice (glaciers), and sea waves .
Process	It is a <i>preparatory</i> process.	It is a <i>transport</i> process.
Example	A rock in the desert cracking under the hot sun (exfoliation).	A river carrying sand and silt downstream.

1.4 AGENTS OF DENUDATION (EROSION) AND HOW THEY SHAPE THE LANDSCAPE

1. Running Water (Fluvial Processes)

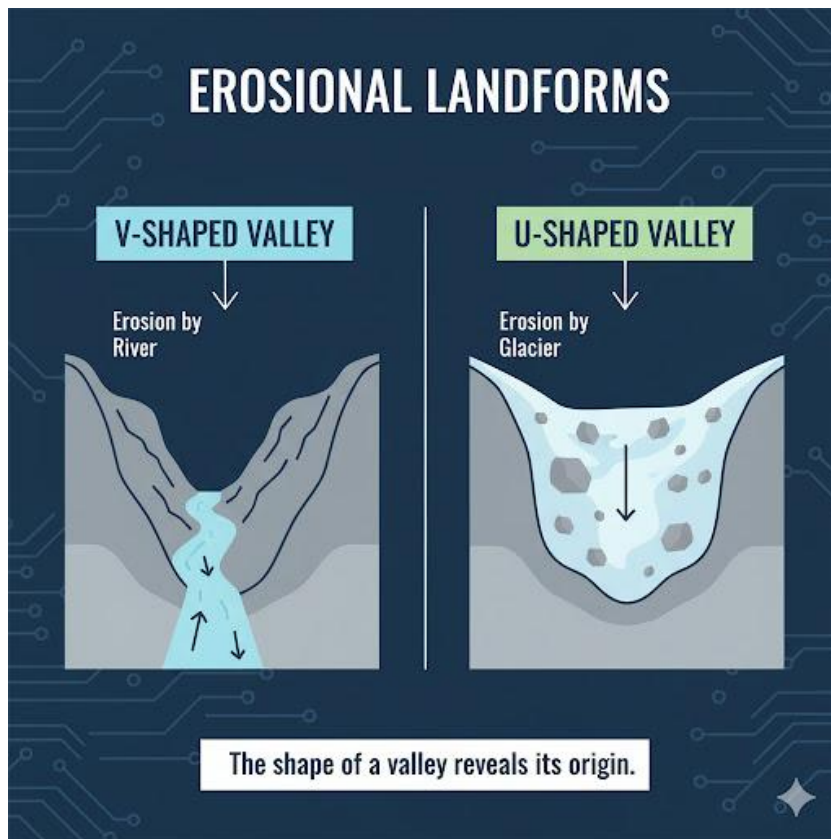
This is the most widespread and dominant agent of denudation, especially in humid regions like Southern Nigeria.

- **Erosional Processes:**

- **Hydraulic Action:** The sheer force of the moving water hitting the river bed and banks, dislodging material.
- **Abrasion (or Corrasion):** The "sandpaper" effect. The river uses its *load* (the rocks and sand it is already carrying) to scrape and grind against the bed and banks.
- **Attrition:** Rocks and pebbles within the river's load collide with each other, breaking down into smaller, smoother, and more rounded particles.
- **Solution (or Corrosion):** The chemical action of the river water dissolving minerals (like limestone) from the rocks.

- **Landforms Created:**

- **Erosional:** V-shaped valleys, waterfalls, rapids, potholes, meanders, river cliffs, gorges.
- **Depositional:** Floodplains, levees, deltas (e.g., the Niger Delta), ox-bow lakes, alluvial fans.



2. Wind (Aeolian Processes)

This is the dominant agent in arid and semi-arid regions (e.g., Northern Nigeria, the Sahel).

- **Erosional Processes:**

- **Deflation:** The process of wind "lifting" and blowing away loose, unconsolidated materials (dust and sand), leaving behind a stony surface (desert pavement).
- **Abrasion:** Wind-blown sand particles sandblast rock surfaces, carving them into new shapes.

- **Landforms Created:**

- **Erosional:** Rock pedestals (a mushroom-shaped rock), Zeugens, Yardangs, Deflation hollows.
- **Depositional:** Sand dunes (barchans, seifs), Loess (fine dust deposited far away).

3. Moving Ice (Glaciers)

This is dominant in high-latitude (polar) and high-altitude (mountain) regions.

- **Erosional Processes:**

- **Plucking:** The glacier freezes onto rock, and as it moves, it "plucks" large blocks of rock from the valley floor.
- **Abrasion:** The glacier uses the rocks embedded in its base to scrape and polish the valley floor, leaving striations (scratches).

- **Landforms Created:**

- **Erosional:** U-shaped valleys, hanging valleys, cirques (armchair-shaped hollows), arêtes (sharp ridges), fjords (drowned glacial valleys).
- **Depositional:** Moraines (ridges of glacial debris), drumlins, outwash plains.

4. Waves (Marine Processes)

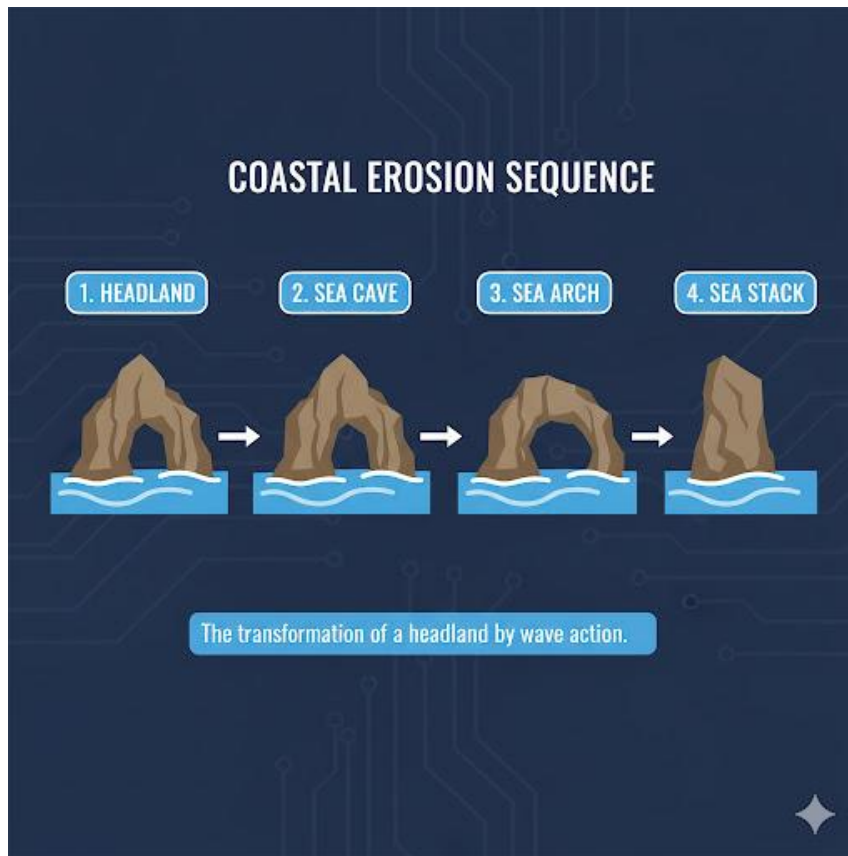
This action is confined to coastlines (e.g., the Lagos and Calabar coast).

- **Erosional Processes:**

- **Hydraulic Action:** The sheer force of waves crashing against a cliff, compressing air in cracks.
- **Abrasion:** Waves hurling shingle and sand against the cliff face.

- **Landforms Created:**

- **Erosional:** Cliffs, wave-cut platforms, headlands, bays, sea caves, sea arches, sea stacks (e.g., the "Bar Beach" erosion).
- **Depositional:** Beaches, spits (e.g., Lagos Bar Beach), bars, tombolos.



1.5 CLIMATIC AND VEGETATION INFLUENCES

- **Climate:** Climate is the single most important factor.
 - **Hot, Wet (Humid) Climate (e.g., Southern Nigeria):** Dominant process is **Chemical Weathering**. Heavy rainfall creates powerful rivers, making **fluvial erosion** the main denudation agent.
 - **Hot, Dry (Arid) Climate (e.g., Northern Nigeria):** Dominant process is **Mechanical Weathering** (e.g., exfoliation). Lack of vegetation and water means **wind erosion** is the main agent.
 - **Cold, (Peri)Glacial Climate:** Dominant process is **Mechanical Weathering** (frost action) and **glacial erosion**.
- **Vegetation:**
 - **Protective Role:** Vegetation (especially forests) is a *barrier* to denudation.
 - Plant roots bind the soil together, preventing erosion.
 - The tree canopy intercepts heavy rain (reducing "splash erosion").

- It acts as a windbreak, reducing wind erosion.
- **Destructive Role:** Plant roots can grow into cracks, widening them (biological weathering).

1.6 HUMAN ACTIVITIES THAT ACCELERATE DENUDATION

Human activities often remove the protective vegetation cover, dramatically accelerating the rate of denudation and leading to environmental degradation.

1. **Deforestation:** Cutting down forests for timber, fuel, or farmland. This exposes the bare soil to rain and wind.
2. **Improper Agricultural Practices:**
 - **Overgrazing:** Allowing too many animals (cattle, goats) to eat all the grass, leaving the soil bare.
 - **Ploughing along slopes:** This creates channels for water to flow, which quickly turn into massive **gullies**. (e.g., The Agulu-Nanka gully erosion in Anambra State).
3. **Mining and Quarrying:** Excavation and digging expose loose rock and soil to erosion.
4. **Construction:** Clearing land for roads, buildings, and cities alters natural slopes, compacts the soil, and creates artificial channels for water, increasing runoff and erosion.

1.7 CONSERVATION AND MANAGEMENT

- **Afforestation/Reforestation:** Planting trees to stabilize soil.
- **Contour Ploughing:** Ploughing *across* a slope (not down) to create small "dams" that trap water.
- **Terracing:** Building "steps" into a steep hillside to create flat areas for farming.
- **Check Dams:** Building small dams inside gullies to slow the water and trap sediment.
- **Cover Cropping:** Planting crops (like *Pueraria*) that cover the soil and protect it.

EVALUATION (WEEK 1)

1. Define denudation and list its three main components.
2. Differentiate between weathering and erosion in a table, using two key aspects.

3. Describe the four main processes of fluvial (river) erosion.
4. Explain how a "Net House" (under Greenhouse Technology) contributes to pest and disease control.
5. Discuss three human activities that accelerate denudation, using specific Nigerian examples if possible.

ASSIGNMENT (WEEK 1)

1. With the aid of diagrams, describe any two landforms created by: (a) Glacial erosion (b) Wave (marine) erosion
2. With examples, explain the role of vegetation in *both* accelerating and controlling the rate of denudation.

WEEK 2: WEATHERING

2.1 MEANING OF WEATHERING

Weathering is the *in-situ* (in place) breakdown, disintegration (physical breaking), or decomposition (chemical rotting) of rocks and minerals at or near the Earth's surface.

It is a **static process**—it does not involve any movement or transportation of the broken material. It is the "preparation" phase for erosion.

2.2 TYPES OF WEATHERING

1. MECHANICAL (PHYSICAL) WEATHERING

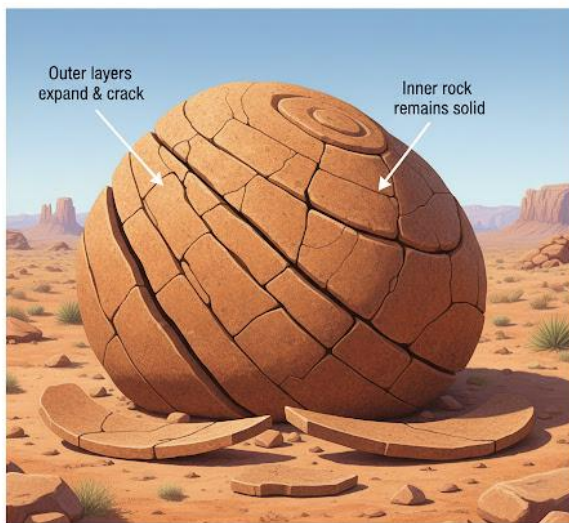
This involves the breakdown of rocks into smaller pieces (known as *regolith*) without any change in their chemical or mineral composition. It is dominant in hot, arid (desert) or very cold (polar) climates.

Processes in Mechanical Weathering:

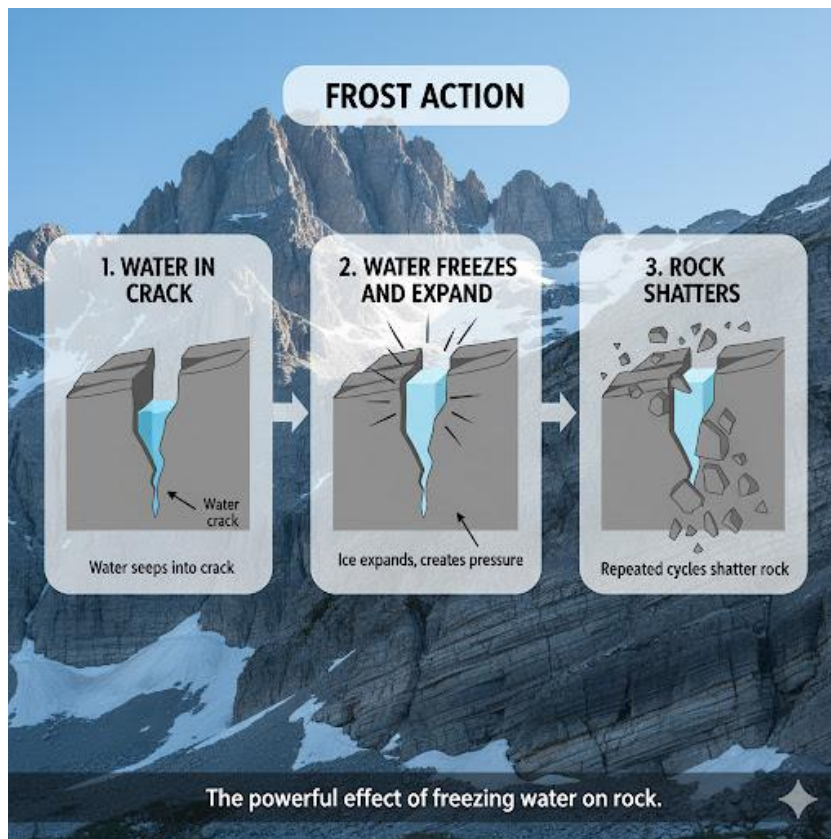
- **Insolation Weathering (Thermal Expansion / Exfoliation):**

- **Process:** In hot deserts, rocks are non-conductors. During the day, the outer surface of the rock heats up and *expands*. At night, the temperature plummets, and the outer surface cools and *contracts*. This repeated expansion and contraction creates stress, causing the outer layers to crack and "peel" off like an onion skin. This specific peeling is called **exfoliation**.
- **Example:** A common process on large, exposed rock domes (inselbergs) like **Zuma Rock** (Niger) or **Olumo Rock** (Ogun).

EXFOLIATION / ONION-SKIN WEATHERING



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- **Frost Action (Freeze-Thaw):**
 - **Process:** Dominant in cold, mountainous regions.
 1. Water seeps into cracks (joints) in a rock.
 2. At night, the water *freezes* and expands by about 9-10% in volume.
 3. This expansion acts like a powerful wedge, widening the crack.
 4. Repeated cycles of freezing and thawing cause the rock to shatter.
 - **Result:** A field of sharp, angular rocks called "scree" at the base of a slope.



- **Salt Crystal Growth:**

- **Process:** In arid or coastal areas, water containing dissolved salts seeps into the pores of rocks. The water evaporates, leaving the salt crystals behind. These crystals grow over time, exerting pressure on the rock pores and causing the rock to disintegrate.

- **Pressure Release (Unloading):**

- **Process:** Deeply buried rocks (like granite) are under immense pressure. When erosion removes the thousands of tonnes of rock above them, the "pressure is released." This causes the rock to expand, crack, and break apart in large sheets.

2. CHEMICAL WEATHERING

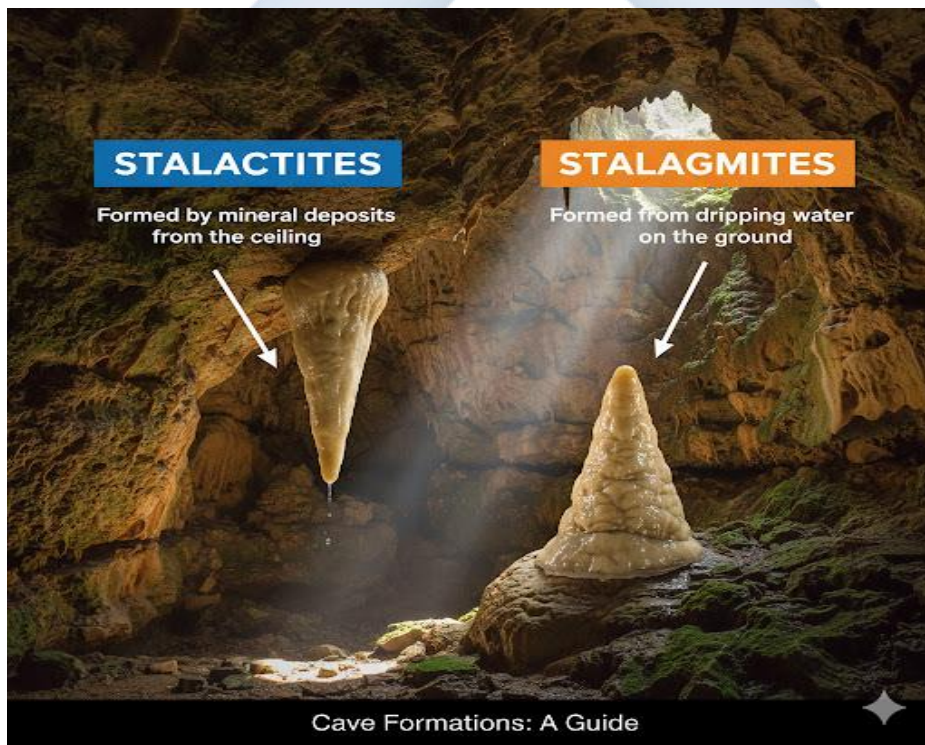
This involves the decomposition of rocks through chemical reactions that change the minerals. It is dominant in **hot, humid (wet) climates** (like Southern Nigeria) because:

1. **Water** (the main agent) is abundant.

2. **Heat** (Van't Hoff's rule) speeds up the rate of all chemical reactions.

Processes in Chemical Weathering:

- **Solution:** The process where water (a universal solvent) dissolves certain minerals.
 - **Example:** Rock salt (halite) or gypsum dissolve completely.
- **Carbonation:**
 - **Process:** Rainwater (H_2O) dissolves carbon dioxide (CO_2) from the air, forming a weak **carbonic acid** (H_2CO_3).
 - This acid is very effective at *dissolving* rocks that contain calcium carbonate, such as **limestone** and **chalk**.
 - **Result:** This process creates "karst" landscapes, including **limestone pavements**, **caves**, **sinkholes**, **stalactites**, and **stalagmites**.



- **Hydrolysis:**

- **Process:** A chemical reaction between water (specifically its H⁺ and OH⁻ ions) and minerals in the rock. This is the *most important* weathering process for silicate rocks (like **granite**).
- **Example:** Hydrolysis breaks down **Feldspar** (a hard mineral in granite) and converts it into **Kaolin** (soft clay). This weakens the entire rock, allowing it to crumble.
- **Oxidation:**
 - **Process:** The reaction of minerals (especially **iron**) with oxygen, usually in the presence of water. It is "rusting."
 - **Example:** Iron compounds in a rock "rust," turning the rock a reddish-brown colour and making it crumble. This is what gives laterite soils their characteristic red colour.
- **Hydration:** The absorption of water *into* a mineral's structure, causing it to expand and weaken (e.g., anhydrite + water → gypsum).

3. BIOLOGICAL WEATHERING

This involves the breakdown of rocks by living organisms.

- **Root Action (Mechanical):** Plant roots (especially tree roots) grow into cracks in rocks. As the root thickens, it acts as a powerful wedge, forcing the rock apart.



- **Burrowing Animals:** Animals like rabbits, rodents, ants, and termites burrow into the ground, moving material and exposing fresh rock surfaces to other forms of weathering.
- **Chemical Action:**
 - **Lichen and Moss:** These organisms grow on bare rock. They produce weak *organic acids* (chelation) that slowly dissolve the minerals, breaking down the rock surface.
 - **Man:** Human activities like mining, quarrying, and construction are a major (and very fast) form of biological weathering.

2.3 FACTORS AFFECTING WEATHERING

1. Climate (The Most Important):

- **Hot/Wet (Humid Tropical):** High temperatures and high rainfall = **Deep, rapid Chemical Weathering** (e.g., hydrolysis, oxidation).
- **Hot/Dry (Arid):** High temperatures, low rainfall = **Strong Mechanical Weathering** (e.g., exfoliation).

- **Cold/Wet (Temperate/Polar):** Cold, fluctuating temperatures = **Strong Mechanical Weathering** (e.g., frost action).
2. **Rock Type (Geology):**
- **Rock Structure:** Rocks with many joints (cracks) or layers will weather faster.
 - **Mineral Composition:** Rocks with "soft" or "reactive" minerals (like limestone, which dissolves) will weather faster than rocks with "hard" or "resistant" minerals (like quartz).
3. **Topography (Relief):**
- **Steep Slopes:** Weathered material (regolith) is removed quickly by gravity, exposing fresh rock to more weathering.
 - **Gentle Slopes:** Allow water to collect and regolith to accumulate, leading to deeper chemical weathering.
4. **Vegetation:** Vegetation (as seen in biological weathering) can both protect the rock (by covering it) and break it apart (with roots and acids).
5. **Time:** The longer a rock is exposed to the elements, the more it will weather.

2.4 WEATHERING AND SOIL FORMATION

Weathering is the *first and most essential step* in creating soil.

- **Role:** Weathering breaks down the **parent rock** (the "bedrock") into smaller mineral particles called **regolith**.
- This regolith (which is a mix of sand, silt, and clay) forms the *inorganic base* of the soil.
- When this regolith is mixed with **organic matter** (decaying plants and animals), **air**, and **water**, it becomes fertile **soil**.
- Mechanical Weathering → produces sandy, coarse soils.
- Chemical Weathering → produces fine-grained, clay-rich soils.

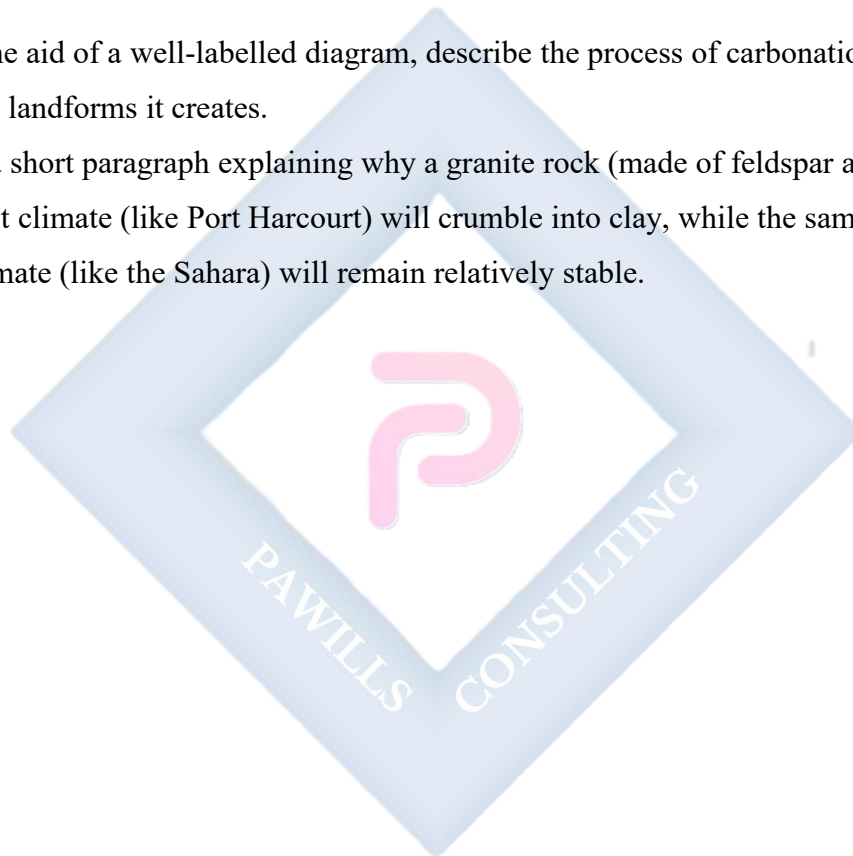
EVALUATION (WEEK 2)

1. (a) Define weathering. (b) Differentiate between mechanical and chemical weathering, giving one process for each.

2. Explain the process of "Exfoliation" (Insolation Weathering) and name a Nigerian landform where it is common.
3. Which type of weathering (mechanical or chemical) is dominant in Southern Nigeria, and why?
4. Describe two (2) ways living organisms (plants/animals) can cause weathering.
5. Briefly explain the role of weathering in soil formation.

ASSIGNMENT (WEEK 2)

1. With the aid of a well-labelled diagram, describe the process of carbonation and name two (2) landforms it creates.
2. Write a short paragraph explaining why a granite rock (made of feldspar and quartz) in a hot, wet climate (like Port Harcourt) will crumble into clay, while the same rock in a hot, dry climate (like the Sahara) will remain relatively stable.



WEEK 3: MASS MOVEMENTS

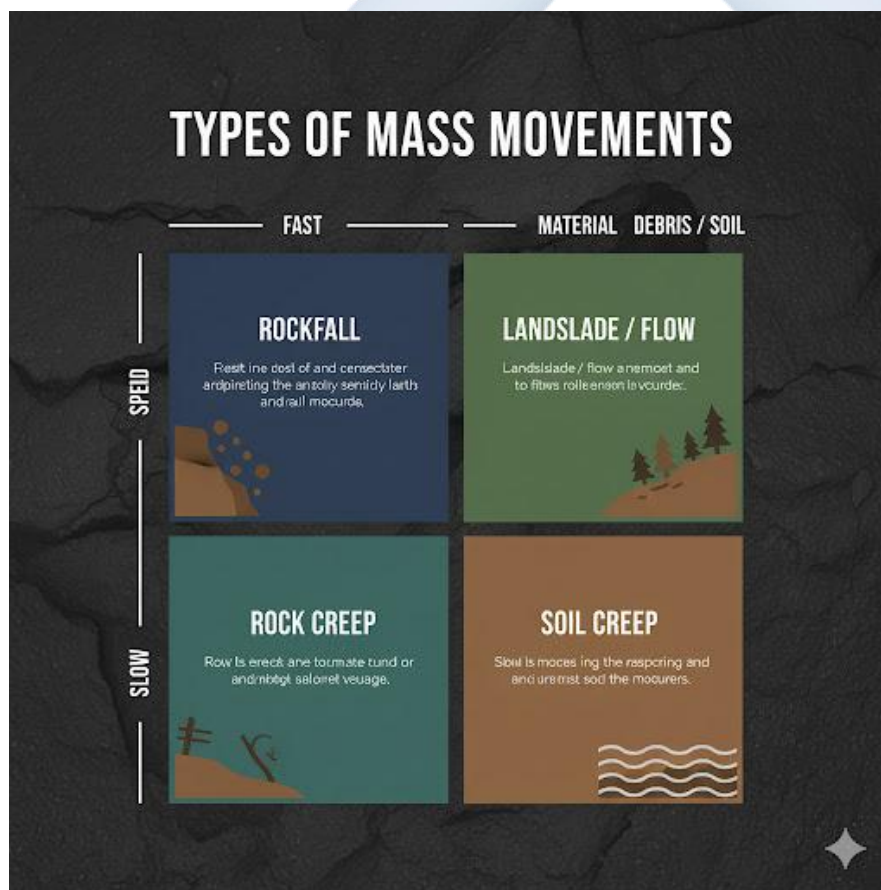
3.1 DEFINITION OF MASS MOVEMENTS (MASS WASTING)

Mass Movement (or Mass Wasting) is the downslope movement of a large mass of rock, soil, and debris (called *regolith*) under the direct influence of **gravity**.

Key Distinction: Unlike erosion, mass movement does *not* require a "transporting agent" like running water, wind, or ice. Gravity alone is the primary force, though water often acts as a *trigger* by lubricating the material.

3.2 CATEGORIES AND TYPES OF MASS MOVEMENTS

Mass movements are classified by their **speed**, **material**, and the **type of motion**.

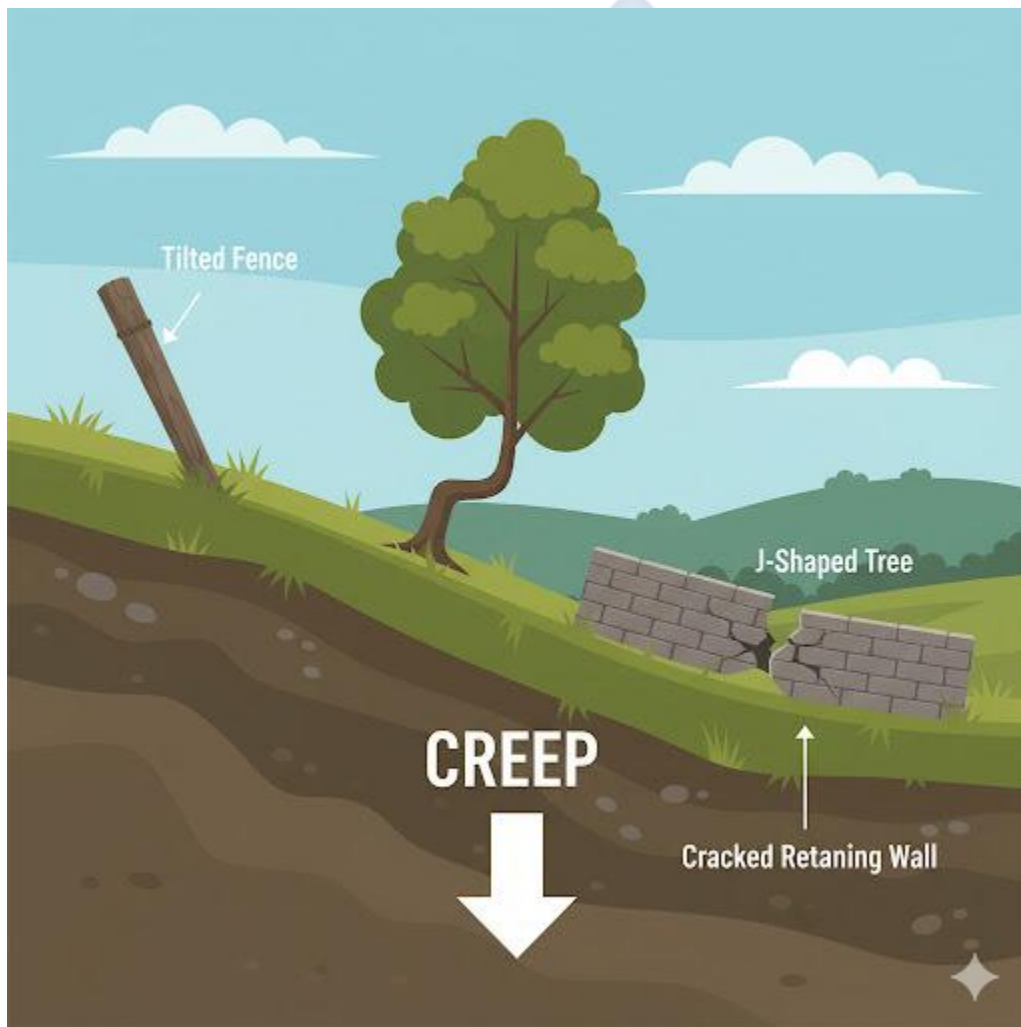


1. SLOW MOVEMENTS

- **CREEP (Soil Creep):**

- **Description:** The *slowest* (imperceptible) and most widespread form of mass movement. It is the gradual, downhill movement of soil and regolith.
- **Process:** Caused by cycles of wetting/drying or freezing/thawing, which cause soil particles to lift and then resettle slightly downhill.

Evidence (You can't see it happen, but you see its effects): Tilted trees ("J-shaped" trunks), bent fences, tilted utility poles, and cracks in roads.



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- **SOLIFLUCTION:**

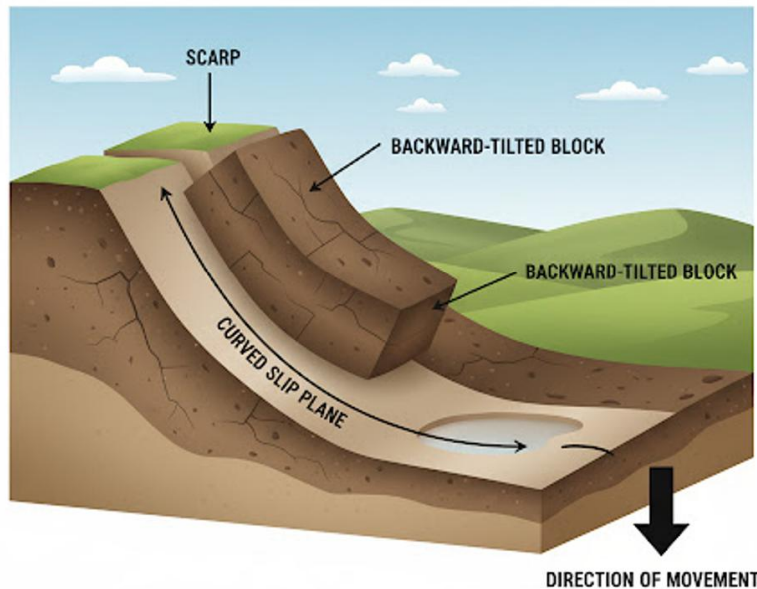
- **Description:** A specific type of creep in cold, sub-polar regions.

- **Process:** In summer, the top layer of frozen ground (permafrost) thaws, but the water cannot drain because of the frozen ice below. The waterlogged, thawed layer becomes a "mush" and flows slowly downslope.

2. FAST MOVEMENTS

- **FLOWS:** The material moves like a viscous *fluid*.
 - **Earthflow:** A mass of earth (soil) flows slowly downslope, often leaving a "tongue-shaped" lobe.
 - **Mudflow:** A rapid (fast) flow of mud and water, common in arid regions after a sudden, intense rainstorm. They follow river channels.
 - **Debris Flow / Debris Avalanche:** A very fast, destructive flow containing a mix of water, mud, rocks, and trees.
- **SLIDES:** The material moves as a *coherent block* along a *defined flat surface* (a "slip plane").
 - **Rockslide:** A block of bedrock detaches and slides rapidly down a flat joint or bedding plane.
 - **Debris Slide:** A slide of unconsolidated soil and rock fragments.
- **SLUMPS (Rotational Slides):**
 - **Description:** A common type of slide where a mass of soil or rock moves *downward and outward* along a *curved* (spoon-shaped) slip surface.
 - **Result:** The top of the slump "rotates" backward, leaving a crescent-shaped scarp (cliff) at the top and a lumpy "toe" at the bottom.

SLUMP: MASS MOVEMENT



- **FALLS:**

- **Description:** The *fastest* type of mass movement. It is the free-fall of detached rock fragments from a steep cliff.
- **Rockfall:** Individual rocks fall and accumulate at the bottom as a "talus" or "scree" slope.

3.3 FACTORS CONTRIBUTING TO MASS MOVEMENTS

These factors can be **pre-conditioning** (what makes the slope unstable) or **triggering** (the specific event that starts the movement).

A. PHYSICAL FACTORS

1. **Slope Gradient (Steepness):** This is the main factor. Gravity has a stronger "pull" on steeper slopes.
2. **Water Content (The #1 Trigger):** Water is the primary trigger.
 - It *lubricates* the soil particles, reducing friction.
 - It *adds weight* to the slope.

- It *increases pore water pressure*, which pushes soil particles apart and "floats" the slope, drastically reducing its strength.
- 3. **Rock Type and Structure:**
 - **Rock Type:** Weak, unconsolidated materials (like clay, shale, sand) are far more likely to fail than hard, crystalline rock (like granite).
 - **Rock Structure:** Layers (bedding planes) that *dip parallel* to the slope, or the presence of joints and faults, create "planes of weakness" that are prone to sliding.
- 4. **Vegetation Cover (Lack of):** Plant roots (especially from trees) *anchor* the soil, holding it together. Removing vegetation makes a slope *much* more unstable.
- 5. **Natural Triggers:**
 - **Earthquakes:** The shaking can instantly trigger massive landslides and rockfalls.
 - **Volcanic Eruptions:** Can melt ice and create mudflows (lahars).

B. HUMAN ACTIVITIES (TRIGGERS / EXACERBATION)

Human activity is a major cause of slope failure.

1. **Deforestation:** As discussed in Week 1, clearing forests on slopes for farming or logging is a primary cause of landslides.
2. **Construction and Excavation:**
 - **Road Cuts:** Cutting into the "toe" (bottom) of a slope to build a road removes its support, often causing the material above to slide.
 - **Loading:** Piling excavated material on the *top* of a slope, or building a heavy house, adds weight and increases the driving force.
3. **Mining and Quarrying:** Blasting and excavation create steep, unstable slopes.
4. **Water Leakage:** Water leaking from irrigation ditches, broken pipes, or septic tanks can saturate a slope, triggering a failure.

3.4 IMPACTS OF MASS MOVEMENTS

- **Social:** Loss of life and property; displacement of communities.
- **Economic:** Destruction of infrastructure (roads, bridges, buildings), loss of valuable farmland, high cost of repair and mitigation.

- **Environmental:** Clogging of rivers with sediment (damaging aquatic life), scarring of the landscape, creation of "landslide dams" (which can later burst and cause flash floods).

3.5 CONTROL AND PREVENTION

1. **Slope Grading:** Re-shaping the slope to a less steep, more stable angle.
2. **Drainage Control:** Installing drains (pipes, channels) to *remove water* from the slope, reducing pore pressure.
3. **Retaining Structures:** Building strong **retaining walls**, **gabions** (wire cages filled with rocks), or **terraces** to hold the slope in place.
4. **Vegetation (Re-afforestation):** Planting trees, grasses, and shrubs. The *roots* anchor the soil. This is the cheapest and most sustainable long-term solution.
5. **Land-Use Planning:** Using GIS (Week 11) to map hazardous areas and *prohibiting* construction on unstable slopes.

EVALUATION (WEEK 3)

1. Define Mass Movement. What is the primary *force* responsible for it, and what is the primary *trigger*?
2. Differentiate between a "Slide" and a "Flow" in terms of material and motion.
3. What are the visible "signs" that slow "Soil Creep" is occurring on a hillside?
4. Explain how a "road cut" (a human activity) can trigger a landslide.
5. List three (3) engineering or biological methods used to control and prevent mass movements.

ASSIGNMENT (WEEK 3)

1. **Gully erosion** in South-Eastern Nigeria (e.g., Agulu-Nanka) is a severe form of erosion that combines fluvial (water) erosion with mass movement.
2. Write a one-page report explaining how gully erosion is a type of mass movement. (Hint: Which type(s) is it? Slumping? Flow?)
3. What *physical* factors (e.g., soil type, rainfall) and *human* factors (e.g., land use) have caused this problem?

WEEK 4: CLIMATIC CHANGE

4.1 MEANING OF CLIMATE CHANGE

Climate Change refers to a significant, long-term, and persistent alteration in the average statistical patterns of weather for a region, or for the Earth as a whole.

- This is *not* the same as **Weather**, which is the short-term state of the atmosphere (e.g., "It is raining today").
- This is also *not* the same as **Climate Variability**, which is a natural fluctuation (e.g., El Niño, a 3-year drought).
- **Climate Change** is a *trend* over decades or centuries, such as the current, rapid **Global Warming** (the increase in Earth's average surface temperature).

4.2 CAUSES OF CLIMATE CHANGE

A. NATURAL CAUSES (HISTORICAL)

The Earth's climate has always changed naturally.

1. **Volcanic Eruptions:** Massive eruptions (e.g., Mount Tambora, 1815) blast huge amounts of **ash** and **sulphur dioxide** into the stratosphere. These particles *reflect* sunlight, leading to a short-term *global cooling*.
2. **Solar Radiation Variations (Sunspots):** The sun's energy output varies in an 11-year "sunspot cycle." This has a minor, but measurable, effect on climate.
3. **Orbital Changes (Milankovitch Cycles):** These are very slow, long-term (thousands of years) changes in the Earth's orbit, tilt, and "wobble." They are the primary driver of the Earth's natural Ice Ages and warm interglacial periods.

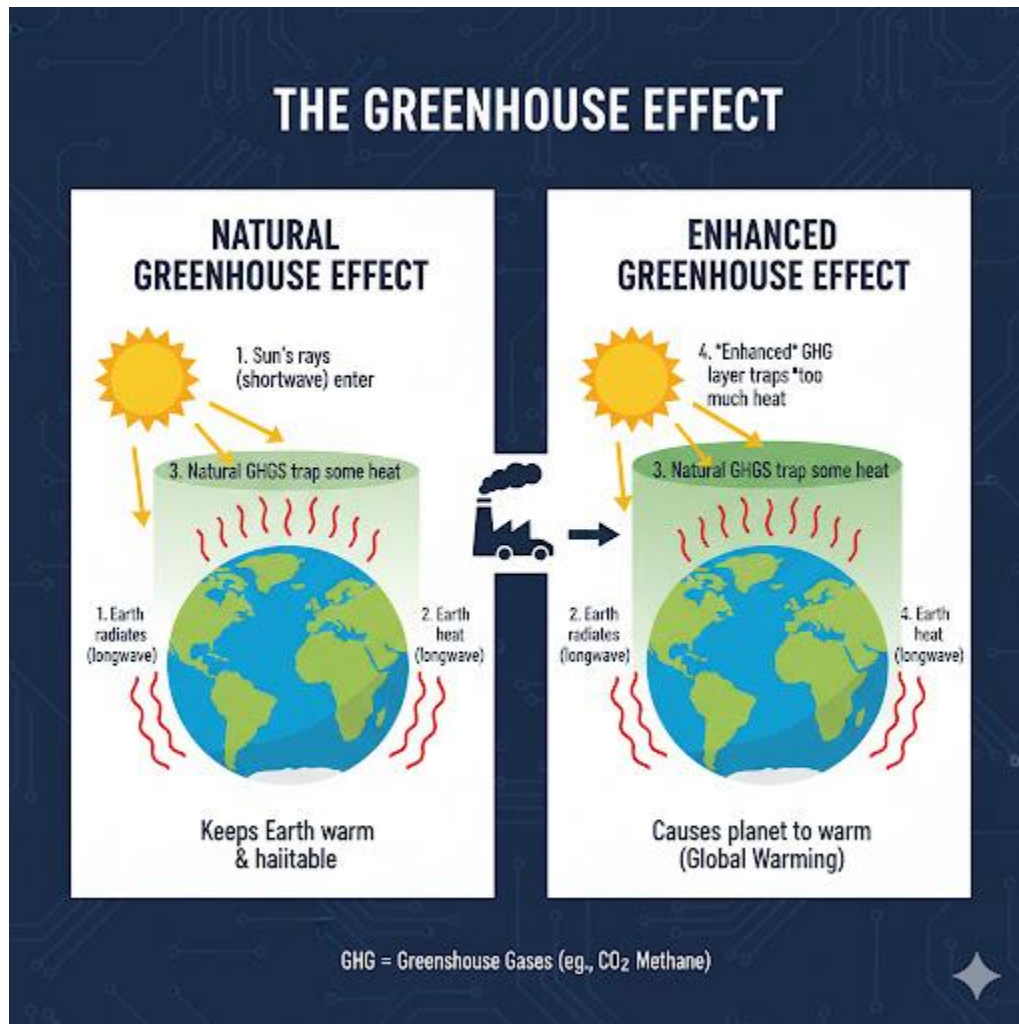
B. ANTHROPOGENIC (HUMAN-INDUCED) CAUSES

The *current, rapid* climate change is overwhelmingly caused by human activities since the Industrial Revolution (c. 1760).

1. The Enhanced Greenhouse Effect:

- **The Greenhouse Effect** is a natural and essential process. Gases in the atmosphere (like CO₂, H₂O) act like a "blanket," trapping some of the sun's heat and keeping the Earth warm enough for life.

The Enhanced Effect is the problem. Human activities are making this "blanket" thicker.



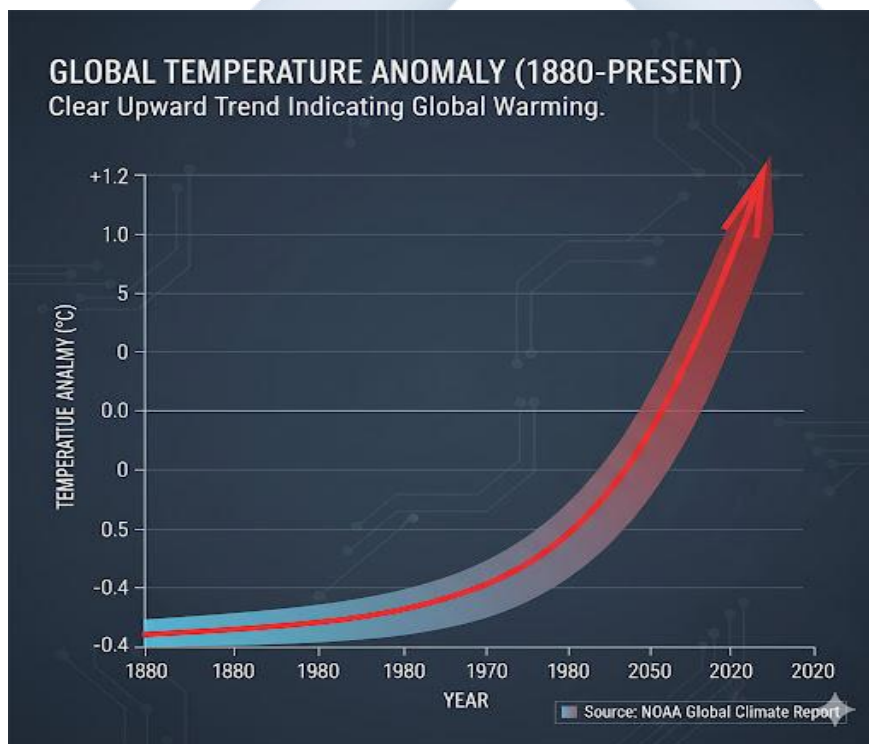
2. Emission of Greenhouse Gases (GHGs):

- **Carbon Dioxide (CO₂):** The main driver. Comes from **burning fossil fuels** (coal, oil, and gas) for electricity, transport (cars, trucks), and industry (cement). Also from **deforestation** (cutting and burning trees).

- **Methane (CH₄):** A more potent, but shorter-lived, GHG. Comes from **agriculture** (e.g., rice paddies, cattle/livestock), **landfills** (rotting waste), and **oil and gas extraction** (gas flaring).
 - **Nitrous Oxide (N₂O):** Comes from **nitrogen-based fertilizers** used in agriculture.
3. **Deforestation:** Forests are "carbon sinks" that absorb CO₂. Cutting them down (1) stops this absorption and (2) releases all the carbon stored in the trees.
 4. **Land-Use Change:** Converting forests to cities or farms changes the "albedo" (reflectivity) of the Earth's surface.

4.3 EVIDENCE OF CLIMATE CHANGE

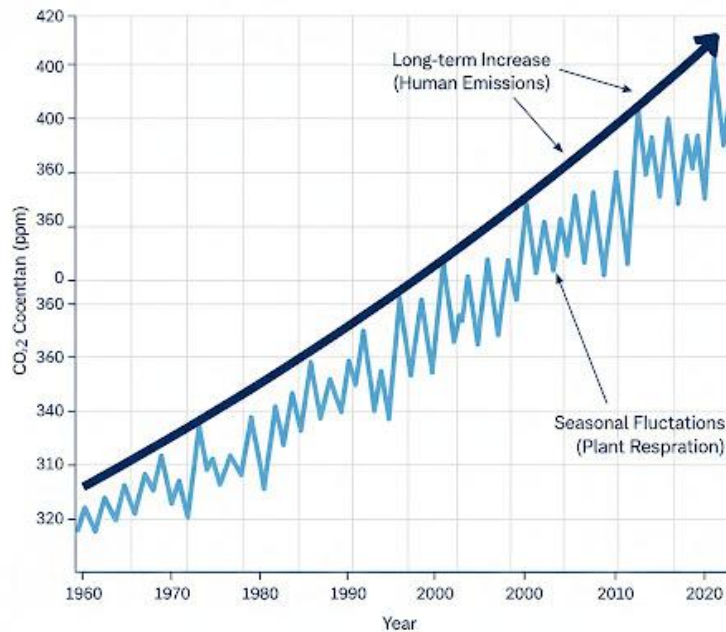
1. **Rising Global Temperatures:** Global average temperatures have risen by over 1.1°C since 1880. The last 10 years (2015-2024) are the warmest decade ever recorded.



2. **Rising Atmospheric CO₂:** Direct measurements (e.g., from Mauna Loa, Hawaii) show CO₂ levels have risen from ~280 parts per million (ppm) pre-industry to over **420 ppm** today, the highest in human history.

The “Keeling Curve”

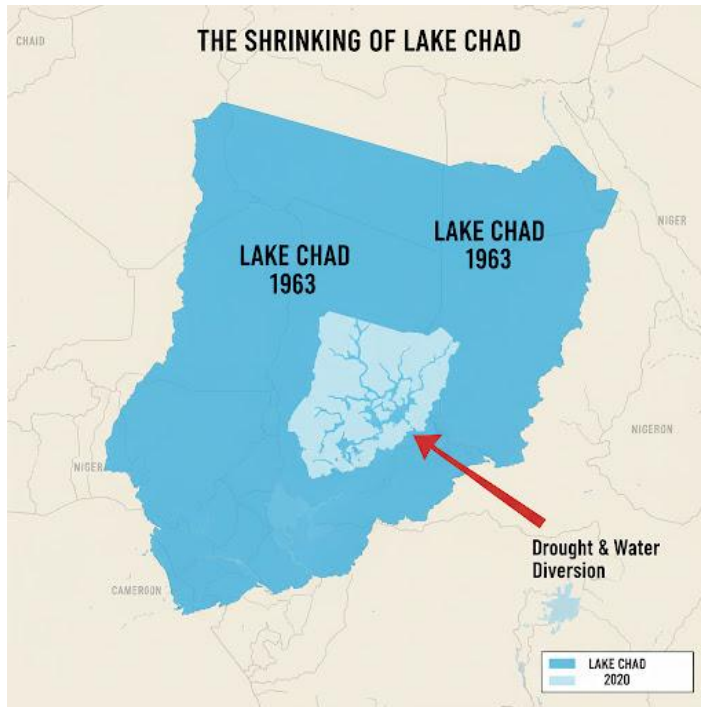
Atmospheric CO₂ Levels at Mauna Loa Observatory



3. **Melting Ice and Glaciers:** Glaciers in mountain ranges (Alps, Himalayas) are rapidly retreating. The Arctic sea ice is shrinking, and the **Greenland and Antarctic ice sheets** are losing mass (melting) at an accelerating rate.
4. **Sea-Level Rise:** The global mean sea level is rising due to:
 - **Thermal Expansion:** Warm water *expands*.
 - **Melting Land Ice:** Water from melting glaciers flows into the sea.
 - This is a critical threat to all coastal cities (like Lagos).

4.4 IMPACTS OF CLIMATE CHANGE (NIGERIA/AFRICA)

1. **Desertification (In the North):**
 - Increased temperatures and more severe droughts are causing the **Sahara Desert to expand southward** ("desertification").
 - **Lake Chad** has shrunk by over 90% since the 1960s, destroying the livelihood of millions in Nigeria, Niger, Chad, and Cameroon.



2. Coastal Flooding (In the South):

- Rising sea levels, combined with storm surges, are causing massive **coastal erosion** and **flooding** in low-lying cities like **Lagos (Victoria Island), Port Harcourt, and Bayelsa.**

3. Farmer-Herder Conflicts (In the Middle Belt):

- This is a key *socio-economic* impact.
- Desertification (Push factor) forces Fulani herders to move *south* in search of pasture and water.
- They (Pull factor) enter the "Middle Belt," where they come into conflict with settled farmers, leading to violence, displacement, and a major threat to food security.

4. Impact on Agriculture:

- More frequent droughts and floods destroy crops.
- Erratic rainfall (e.g., rain coming too late, or too hard) disrupts the planting season.

5. Health Impacts: Higher temperatures can lead to more heatstroke and expand the range of diseases like malaria (carried by mosquitoes).

4.5 MITIGATION AND ADAPTATION

1. **Mitigation (Addressing the Cause):** Actions to *reduce* greenhouse gas emissions.
 - **Global:** The **Paris Agreement (2015)**, where all nations agreed to Nationally Determined Contributions (NDCs) to limit warming.
 - **Local:**
 - Shifting from fossil-fuel generators to **Renewable Energy** (Solar, Wind, Hydro).
 - **Afforestation:** Planting trees (e.g., Nigeria's "Great Green Wall" project in the North).
 - Improving **Energy Efficiency** (e.g., using LED bulbs).
2. **Adaptation (Addressing the Effect):** Actions to *cope* with the impacts that are already happening.
 - **Example:** Building sea walls (like the "Great Wall of Lagos" for Eko Atlantic) to protect against sea-level rise.
 - **Example:** Using solar-powered irrigation (Week 1, Term 2) to adapt to droughts.
 - **Example:** Planting drought-resistant crop varieties.

EVALUATION (WEEK 4)

1. Differentiate clearly between "Weather," "Climate Variability," and "Climate Change."
2. Explain the "Enhanced Greenhouse Effect." What are the two main GHGs humans are releasing?
3. List four (4) major pieces of *evidence* for global climate change.
4. Describe two (2) specific impacts of climate change on Nigeria.
5. What is the difference between "Mitigation" and "Adaptation" as responses to climate change? Give one example of each.

ASSIGNMENT (WEEK 4)

1. Research the "Great Green Wall" initiative in Africa.
2. Write a one-page report on its *aims* and the *challenges* it faces in implementation.

WEEK 5: MIDTERM EXAMINATION

This week is for the Midterm Examination, covering all topics from Weeks 1-4 (Denudation, Weathering, Mass Movements, Climate Change).



WEEK 6: ECOWAS

6.1 INTRODUCTION

The **Economic Community of West African States (ECOWAS)** is a regional economic and political bloc of 15 West African countries. It is one of the "building blocks" of the African Union (AU).

6.2 HISTORICAL BACKGROUND AND ESTABLISHMENT

- **Context:** After independence, West Africa was fragmented along linguistic and colonial lines (Anglophone vs. Francophone). This hindered trade and cooperation.
- **Formation:** Led by Nigeria (under Yakubu Gowon) and Togo (under Gnassingbé Eyadéma), the 15 states signed the **Treaty of Lagos** on **May 28, 1975**.
- **Founding Member States:** Benin, Burkina Faso, Côte d'Ivoire, Gambia, Ghana, Guinea, Guinea-Bissau, Liberia, Mali, Mauritania (withdrew in 2000), Niger, Nigeria, Senegal, Sierra Leone, Togo. (Cape Verde joined later).
- **Headquarters:** Abuja, Nigeria.

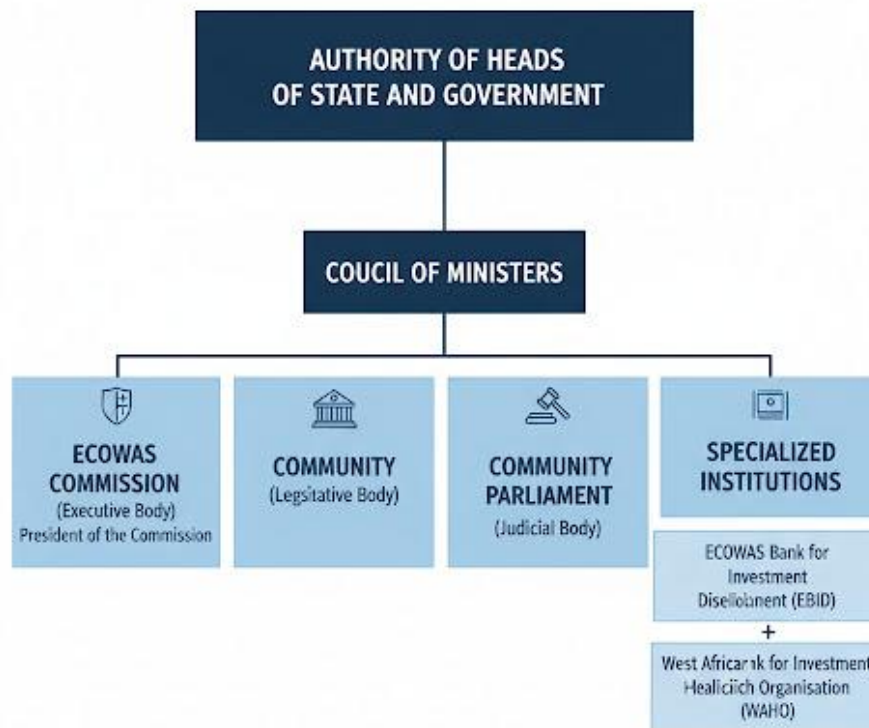
6.3 OBJECTIVES OF ECOWAS

The primary goal is to promote economic cooperation and integration.

1. **Economic Integration:** To create a **Common Market**, removing tariffs and trade barriers (via the **Trade Liberalization Scheme - ETLS**) and establishing a **Common External Tariff (CET)**. The ultimate (but difficult) goal is a single currency (**Eco**).
2. **Free Movement:** To allow the free movement of *persons, goods, and services* within the region (leading to the **ECOWAS Passport** and visa-free travel).
3. **Peace and Security:** To maintain regional peace, stability, and security. This has become one of its most important functions.
4. **Regional Cooperation:** To harmonize policies in agriculture, energy (e.g., West African Power Pool - WAPP), and transport.

6.4 STRUCTURE AND ORGANS

ORGANIZATIONAL STRUCTURE OF ECOWAS



1. **Authority of Heads of State and Government:** The *supreme* decision-making body (the 15 Presidents).
2. **Council of Ministers:** The "cabinet," composed of ministers (e.g., Foreign Affairs, Finance) from each state.
3. **ECOWAS Commission:** The "secretariat" or civil service, based in Abuja. It implements decisions and is led by the President of the Commission.
4. **Community Parliament:** A representative body (mostly advisory) that debates regional issues.
5. **Community Court of Justice:** A judicial body that interprets the ECOWAS treaty and, crucially, handles human rights cases.
6. **ECOMOG (ECOWAS Monitoring Group):** The (non-standing) military peacekeeping arm.

6.5 ACHIEVEMENTS

1. **Peacekeeping (ECOMOG):** This is its greatest success. ECOMOG, (though largely funded and staffed by Nigeria) successfully intervened to end brutal civil wars in **Liberia** and **Sierra Leone**. It also intervened to uphold democracy in **The Gambia** (2017).
2. **Free Movement of Persons:** The "Protocol on Free Movement" is a major achievement, allowing citizens to travel, trade, and live in other member states for up to 90 days without a visa.
3. **Democratic Standards:** ECOWAS has developed a "zero-tolerance" policy for military coups. It has (in the past) successfully suspended and sanctioned member states to pressure a return to democracy.

6.6 CHALLENGES

1. **Political Instability:** The *biggest* current challenge. The recent wave of military coups in **Mali, Guinea, Burkina Faso, and Niger** (the "Sahel Coup Belt") has created a major political crisis and undermined the bloc's authority.
2. **Monetary Union Failure:** The goal of launching the single currency, the **Eco**, has been postponed for decades. The Francophone countries (which use the CFA Franc) and Anglophone countries have failed to meet the "convergence criteria."
3. **Non-Payment of Dues:** Many member states fail to pay their membership fees, which starves the organization of funds (except for Nigeria, which pays the largest share).
4. **Over-dominance of Nigeria:** Nigeria's size (over 60% of the region's GDP and population) makes smaller states fear "domination."
5. **Smuggling / Non-Implementation:** Despite the free trade "on paper," many countries use porous borders and corrupt officials to undermine the rules (e.g., smuggling rice, cars, or petrol).

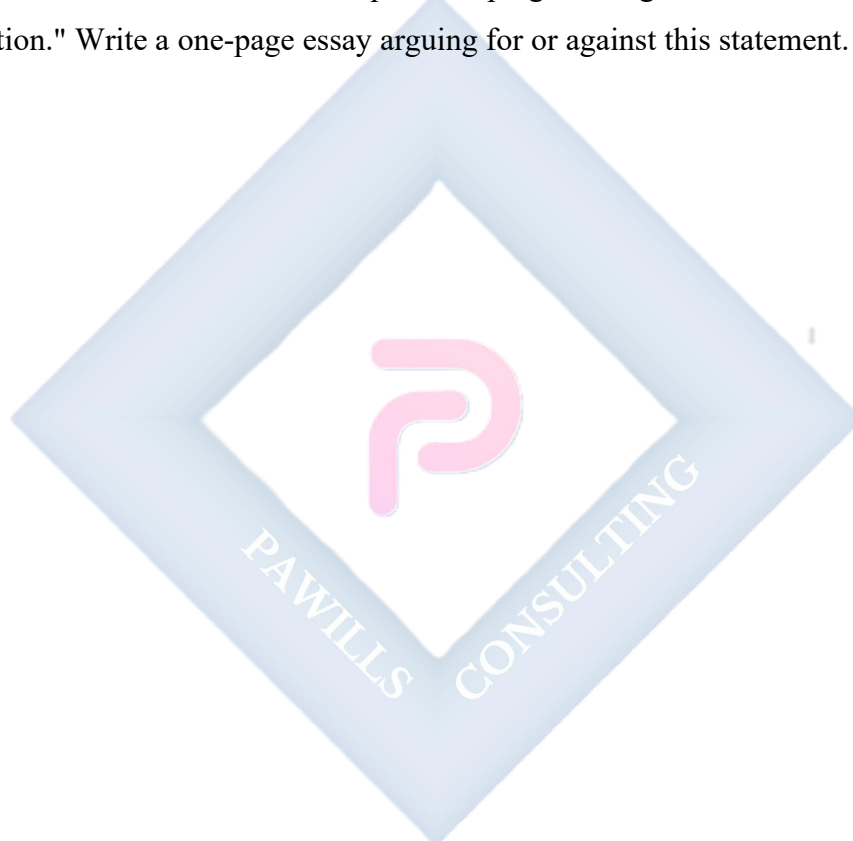
EVALUATION (WEEK 6)

1. When and where was ECOWAS established?

2. What is the **primary** economic objective of ECOWAS?
3. What is the **supreme** organ of ECOWAS?
4. What is **ECOMOG**, and what is considered its greatest achievement?
5. List three (3) major challenges facing ECOWAS today.

ASSIGNMENT (WEEK 6)

1. Draw a map of West Africa and label the 15 ECOWAS member states.
2. "The achievements of ECOWAS in peacekeeping outweigh its failures in economic integration." Write a one-page essay arguing for or against this statement.



WEEK 7: MIDTERM BREAK

This is a break from instruction. Students are encouraged to review the first half of the term's work.



WEEK 8: TRADE

8.1 MEANING OF TRADE

Trade is the fundamental economic activity of **buying, selling, or exchanging goods and services**. It is the engine of an economy, allowing for the movement of goods from producers (who have a surplus) to consumers (who have a need).

8.2 TYPES OF TRADE

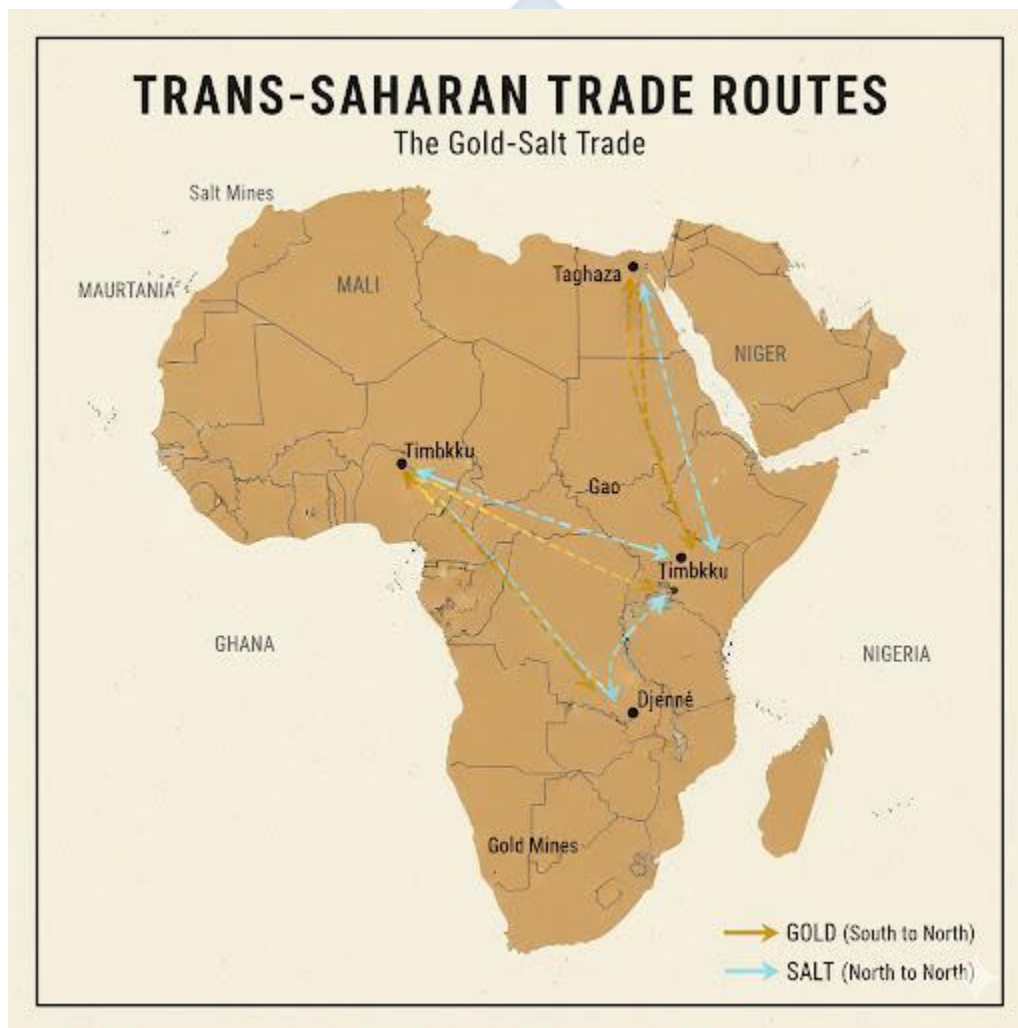
1. **Internal (Domestic) Trade:** Trade *within* the boundaries of a single country.
 - **Wholesale:** Buying in *bulk* from the producer and selling in *smaller bulk* to the retailer.
 - **Retail:** Selling in *small units* directly to the final consumer.
 - *Example:* Cocoa from Ondo State is sold to a factory in Lagos.
2. **External (International) Trade:** Trade *across* national borders.
 - **Import:** Buying goods *from* other countries (e.g., Nigeria buys cars from Japan).
 - **Export:** Selling goods *to* other countries (e.g., Nigeria sells crude oil to India).
 - **Entrepôt:** An "intermediary" trade. (e.g., Singapore imports raw rubber, processes it, and *re-exports* it as tires).
3. **Barter Trade:** The (ancient) direct exchange of goods for other goods, *without* the use of money.

8.3 IMPORTANCE OF TRADE IN ECONOMIC DEVELOPMENT

1. **Source of Foreign Exchange (Forex):** International trade (exports) is the primary way a nation earns foreign currency (e.g., USD) needed to pay for its imports.
2. **Encourages Specialization:** Trade allows a country to *specialize* in what it does best (its "comparative advantage") and import the rest. (e.g., Nigeria specializes in oil, Ghana specializes in cocoa).
3. **Creates Employment:** The entire trade value chain (producers, transporters, bankers, marketers, retailers) creates millions of jobs.

4. **Wider Market & Economies of Scale:** Trade gives producers access to a global market, allowing them to produce in *bulk* (economies of scale), which lowers the cost per item.
5. **Access to Variety & Technology:** Imports allow citizens to access goods, services, and new technologies (e.g., smartphones, computers) that are not available locally.
6. **Fosters International Relations:** Trade builds economic and political links between countries.

8.4 HISTORICAL TRADE ROUTES IN WEST AFRICA



1. Trans-Saharan Trade Routes (Ancient):

- **Who:** West African empires (Ghana, Mali, Songhai) trading with North African Berbers.

- **Route:** Across the Sahara Desert.
- **Key Goods:** The "Gold-for-Salt" trade. West Africa exported **gold, ivory, and slaves** *north*. North Africa imported **salt, cloth, and horses** *south*.
- **Impact:** This trade built the wealth of Ghana, Mali, and Songhai and introduced Islam to West Africa.

2. Coastal Trade Routes (Trans-Atlantic):

- **Who:** Coastal African kingdoms (e.g., Oyo, Benin, Asante) trading with Europeans (Portuguese, British, etc.).
- **Route:** From coastal ports (e.g., Lagos, Badagry, Elmina) across the Atlantic.
- **Key Goods:** Initially gold, ivory, and pepper. This was tragically replaced by the **Trans-Atlantic Slave Trade** (1500s-1800s).
- **Impact:** This trade reoriented the entire West African economy *away* from the interior (Sahara) and *towards* the coast, a pattern that still exists.

8.5 CHALLENGES FACING TRADE IN WEST AFRICA

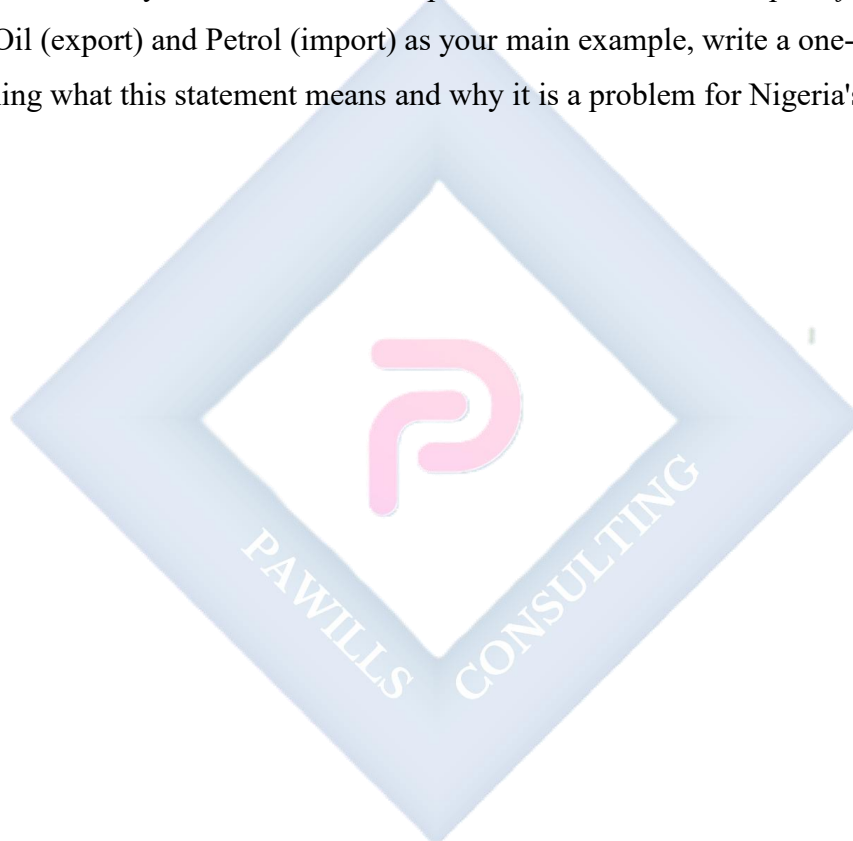
1. **Poor Infrastructure:** This is the *biggest* challenge. Inadequate roads, congested ports (e.g., Apapa), and a lack of railways make it extremely slow and expensive to move goods.
2. **Over-reliance on Primary Products:** Most West African exports are *raw materials* (e.g., cocoa, oil, cashews). We fail to *process* these goods, so we lose the "value-added" profit. (We sell cocoa for \$2,000/tonne and buy chocolate for \$20,000/tonne).
3. **Trade Barriers (Tariffs & Bureaucracy):**
 - Despite ECOWAS, there are still too many tariffs, corrupt officials, and endless "checkpoints" on the roads that harass traders and demand bribes.
4. **Insecurity:** Banditry, piracy (in the Gulf of Guinea), and terrorism (in the Sahel) make trade routes unsafe.
5. **Political Instability:** Coups and border closures (e.g., Nigeria's 2019 land border closure) create uncertainty and disrupt supply chains.

EVALUATION (WEEK 8)

1. Define trade and differentiate between "Internal" and "External" trade.
2. What was the "Gold-for-Salt" trade?
3. What was the main *economic* impact of the Trans-Atlantic trade on West Africa's geography?
4. List three (3) major challenges facing trade in West Africa today.

ASSIGNMENT (WEEK 8)

1. "Nigeria's economy suffers because it exports *raw materials* but imports *finished goods*."
2. Using Oil (export) and Petrol (import) as your main example, write a one-page essay explaining what this statement means and why it is a problem for Nigeria's trade balance.



WEEK 9: TOURISM

9.1 DEFINITION AND KEY COMPONENTS

Tourism is the act and process of spending time away from one's usual environment (for leisure, business, or other purposes) for a period of *not more than one consecutive year*. It is a major global *service industry*.

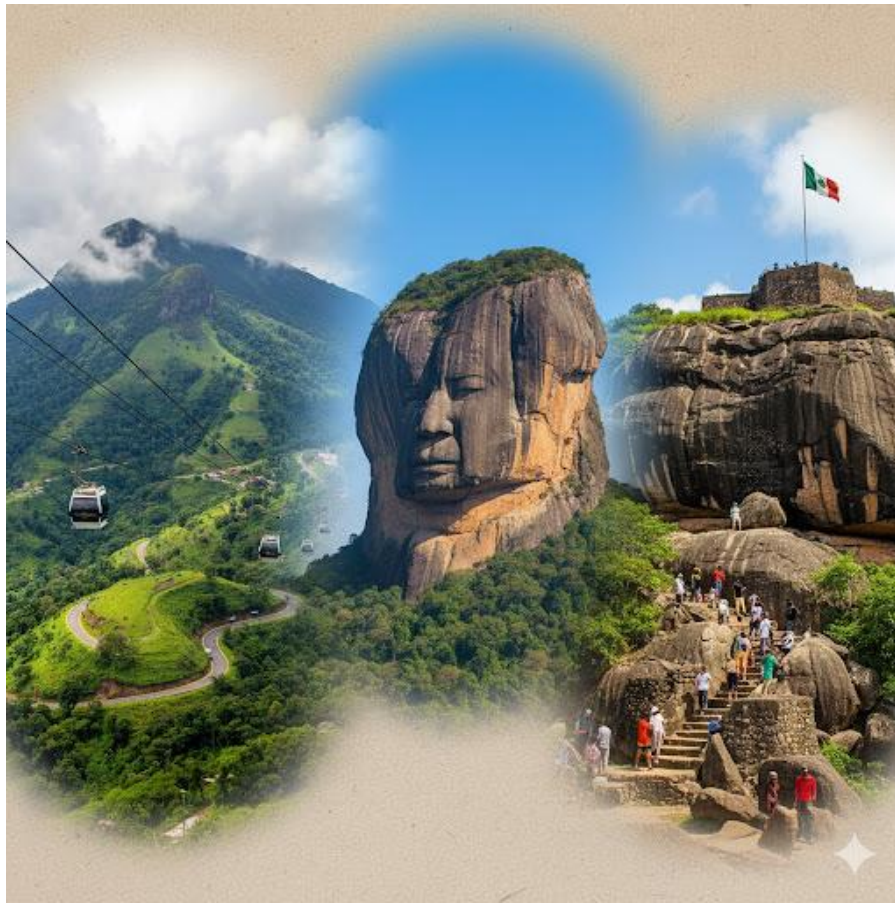
Key Components of the Tourism Industry:

1. **Attractions:** The "pull factor." The *reason* to visit (e.g., natural wonders, cultural sites, festivals, beaches).
2. **Accommodation:** Where tourists stay (e.g., hotels, resorts, guesthouses).
3. **Transport:** How tourists get *to* and *around* the destination (e.g., airlines, airports, roads, tour buses).
4. **Amenities & Support Services:** All other services tourists need (e.g., restaurants, tour guides, banks, security, healthcare).

9.2 TYPES OF TOURISM

1. **Domestic Tourism:** Travel *within* one's own country (e.g., a family from Lagos visiting Yankari Game Reserve).
2. **International Tourism:** Travel *outside* one's country.
3. **Eco-Tourism:** Responsible travel to natural areas that *conserves* the environment and *improves* the well-being of local people (e.g., Obudu Mountain Resort).
4. **Cultural/Heritage Tourism:** Travel to experience the history, art, festivals, and culture of a place (e.g., visiting the Osun-Osogbo Sacred Grove, the Eyo Festival, or Benin City).
5. **Business Tourism (MICE):** Meetings, Incentives, Conferences, and Exhibitions. (e.g., a large conference at the Eko Hotel).
6. **Religious Tourism:** Travel for pilgrimage (e.g., to Mecca, Jerusalem, or to large religious camps in Nigeria).

9.3 MAJOR TOURIST ATTRACTIONS IN NIGERIA



Tourist Attraction	Location	Description
Obudu Mountain Resort	Cross River State	A temperate-climate resort on a plateau. Famous for its beautiful landscapes, “holy mountain,” and cable car system.
Yankari Game Reserve	Bauchi State	Nigeria’s premier wildlife park. Home to elephants, baboons, and (formerly) lions. Famous for the Wikki Warm Springs.
Zuma Rock	Niger State (near Abuja)	A massive inselberg (monolith) with a “human face” on one side. A prominent national symbol.
Olumo Rock	Abeokuta,	A historical rock outcrop used as a fortress and refuge by the

	Ogun State	Egba people during 19th-century wars. It has elevators and stairs for modern tourists.
Osun-Osogbo Sacred Grove	Osun State	A UNESCO World Heritage Site. A protected forest grove on the banks of the Osun River, filled with shrines, sculptures, and art dedicated to the river goddess, Osun.
Erin Ijesha Waterfall	Osun State	A large, multi-level (seven-step) waterfall.
Ogbunike Caves	Anambra State	A large cave system with spiritual and historical significance, also a UNESCO Heritage Site.

9.4 IMPORTANCE OF TOURISM

1. **Generates Foreign Exchange:** International tourists (business and leisure) bring in "hard" currency (USD, EUR).
2. **Creates Employment:** A powerful job creator. It creates jobs *directly* (hotel staff, tour guides) and *indirectly* (farmers who supply the hotel, taxi drivers, craft-makers).
3. **Stimulates Infrastructure:** Governments are encouraged to build better roads, airports, and power systems to support a tourist-friendly environment.
4. **Preserves Culture & Environment:** *Eco-tourism* and *Cultural Tourism* provide a direct *economic incentive* for a community to protect its local forest (like Yankari) or preserve its festival (like Osun-Osogbo), because it is now a source of income.
5. **Boosts Local Economies:** Tourists spend money, which circulates directly into the local community.

9.5 CHALLENGES LIMITING TOURISM IN NIGERIA

1. **Insecurity:** This is the **#1 challenge**. Widespread issues of kidnapping, banditry, and terrorism make travel on roads to many sites extremely dangerous, deterring both local and international tourists.
2. **Poor Infrastructure:** Inadequate roads to remote sites, a lack of consistent electricity (forcing hotels to run on expensive generators), and poor maintenance of facilities.

3. **Poor Management & Marketing:** Many sites are poorly managed. There is also a lack of a strong, positive, and consistent *national brand* to "sell" Nigeria as a destination.
4. **Negative Global Image:** Nigeria's international image is often dominated by news of corruption, scams, and insecurity, which overshadows its beauty.
5. **Lack of Funding:** The government has historically underfunded the tourism ministry and the maintenance of national parks and monuments.

EVALUATION (WEEK 9)

1. Define Tourism and list its four main components.
2. Differentiate between "Domestic Tourism" and "Eco-Tourism."
3. Name three (3) major tourist attractions in Nigeria and their states.
4. Explain two (2) ways tourism can help *preserve* culture and the environment.
5. What is the **number one** challenge limiting the growth of tourism in Nigeria today?

ASSIGNMENT (WEEK 9)

1. You are appointed the new Minister of Tourism for Nigeria.
2. Write a one-page "Action Plan" titled "How to Revive Nigerian Tourism."
3. It must have three sections:
 - (1) How to solve the security problem.
 - (2) How to solve the infrastructure problem.
 - (3) How to solve the "image" problem.

WEEK 10: SATELLITE REMOTE SENSING

10.1 DEFINITION OF REMOTE SENSING

Remote Sensing is the science and art of acquiring information about an object, area, or phenomenon *without* being in direct physical contact with it.

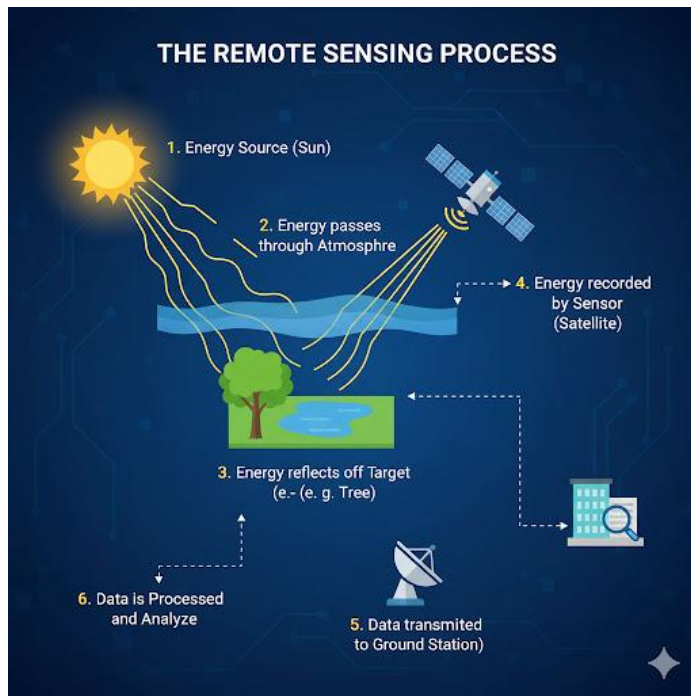
- Your *eyes* are a remote sensor.
- A *camera* is a remote sensor.
- **Satellite Remote Sensing** is the specific process of using **satellites** (platforms) equipped with **sensors** to capture data about the Earth's surface and atmosphere.

10.2 DISTINCTION FROM OTHER DATA METHODS

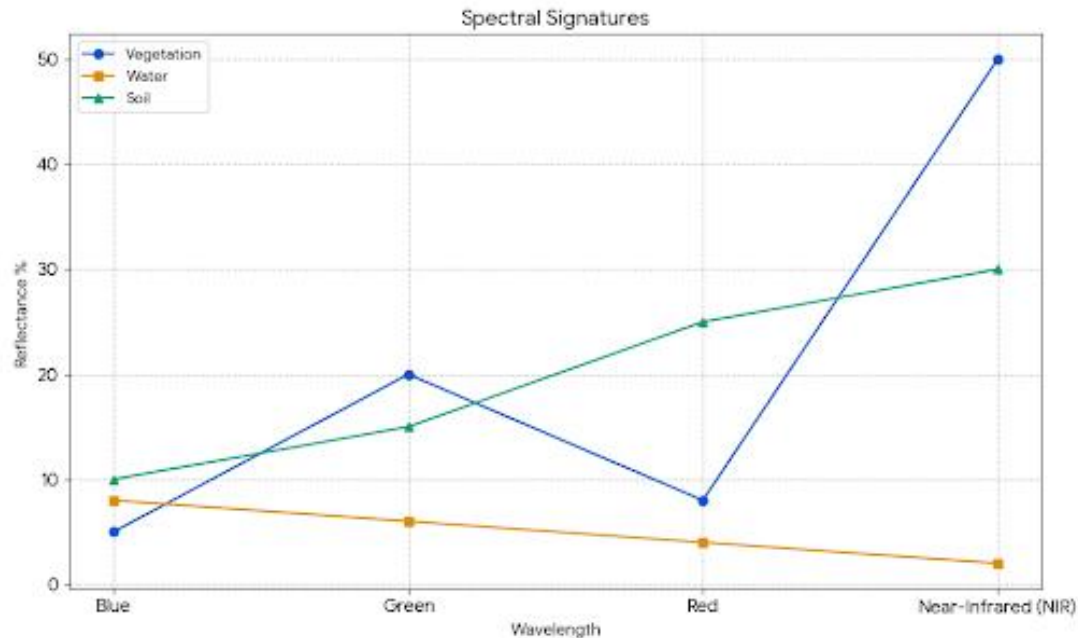
Method	Data Source	Scale
Field Survey	Physical presence (e.g., GPS, soil samples)	Local, very detailed.
Aerial Photography	Low-altitude planes/drones	Local/Regional, high detail.
Satellite Remote Sensing	Space-based satellites	Regional/Global , broad area.

10.3 PRINCIPLES OF REMOTE SENSING

The process works in a simple, standardized way:



- **Spectral Signature:** The key principle. Different objects on Earth's surface (e.g., water, vegetation, soil, buildings) *reflect* and *absorb* energy (light) in unique ways. A sensor can identify an object by this unique "signature."
 - *Example:* Healthy vegetation absorbs red light and strongly reflects **Near-Infrared (NIR)**. A sensor can see this NIR reflection (which our eyes cannot) and instantly map all healthy crops.



This graph illustrates how different materials reflect energy across various wavelengths:

- **Vegetation:** Reflects strongly in the Near-Infrared (NIR) and Green wavelengths, but absorbs Blue and Red light (for photosynthesis).
- **Water:** Reflects most in the Blue wavelength and absorbs energy strongly as wavelengths increase, especially in the NIR.
- **Soil:** Generally shows an increasing reflectance from Blue to NIR, with drier soil being more reflective than wet soil.

10.4 TYPES OF SENSORS

1. Passive Sensors:

- **How:** They are *passive*. They need an external energy source, so they *record* natural energy, which is usually **reflected sunlight**.
- **Example:** Your eyes, a normal camera, **Landsat, Sentinel-2**.
- **Limitation:** **Cannot** see through clouds. **Cannot** work at night.

2. Active Sensors:

- **How:** They are *active*. They *emit* their own energy (a pulse of radiation) and then *record* the "backscatter" (the reflection).
- **Example:**
 - **RADAR (Radio Detection and Ranging):** Sends out radio waves.
 - **LiDAR (Light Detection and Ranging):** Sends out laser pulses.
- **Advantage:** Can work **at night** and (especially RADAR) can "see" **through clouds**.

10.5 TYPES OF RESOLUTION

- **Spatial Resolution:** The *detail* or pixel size. (e.g., **30m** (Landsat - one pixel is 30m x 30m) vs. **1m** (Ikonos - high resolution)).
- **Temporal Resolution:** The *frequency* or "revisit time." (e.g., "Landsat revisits every **16 days**.")

10.6 MAJOR SATELLITES

- **Landsat (USA):** The "workhorse" for 50+ years. Monitors land use, deforestation, agriculture.
- **Sentinel (EU):** A modern series. Sentinel-2 is like Landsat. Sentinel-1 is a RADAR (active) satellite, excellent for flood mapping.
- **SPOT (France):** High-resolution, used for mapping.
- **NigeriaSat-1, NigeriaSat-2, NigeriaSat-X:** Nigeria's own Earth Observation satellites, run by **NASRDA** (National Space Research and Development Agency). They are used for mapping, disaster management, and security.

10.7 APPLICATIONS OF SATELLITE REMOTE SENSING

1. **Agriculture:** Monitoring crop health, predicting drought, and estimating yield.
2. **Environmental Monitoring:**
 - Tracking **deforestation** (e.g., in the Amazon and Nigeria).
 - Tracking **desertification** and the shrinking of **Lake Chad**.
 - Monitoring **oil spills** in the Niger Delta.

3. **Disaster Management:**

- **Flood Mapping:** RADAR sensors can see the water *through* the clouds during a flood, showing exactly which areas are inundated.
- Assessing damage from fires.

4. **Urban Planning:** Mapping urban sprawl (how cities are growing) and land-use change.

5. **Climate Change:** Monitoring sea-level rise and the melting of ice caps.

EVALUATION (WEEK 10)

1. Define Satellite Remote Sensing.
2. What is the "Spectral Signature" of an object?
3. Differentiate *clearly* between a "Passive Sensor" and an "Active Sensor."
4. What is the main *advantage* of an Active (RADAR) Sensor?
5. List three (3) specific applications of remote sensing in Nigeria.

ASSIGNMENT (WEEK 10)

1. Research Nigeria's space agency, **NASRDA**.
2. Write a one-page report on the **NigeriaSat-2** satellite. When was it launched, and what are its main functions?

WEEK 11: GIS APPLICATION

11.1 MEANING OF GIS

A **Geographic Information System (GIS)** is a powerful computer-based system for **capturing, storing, checking, manipulating, analyzing, and displaying (visualizing)** geographically referenced data (spatial data).

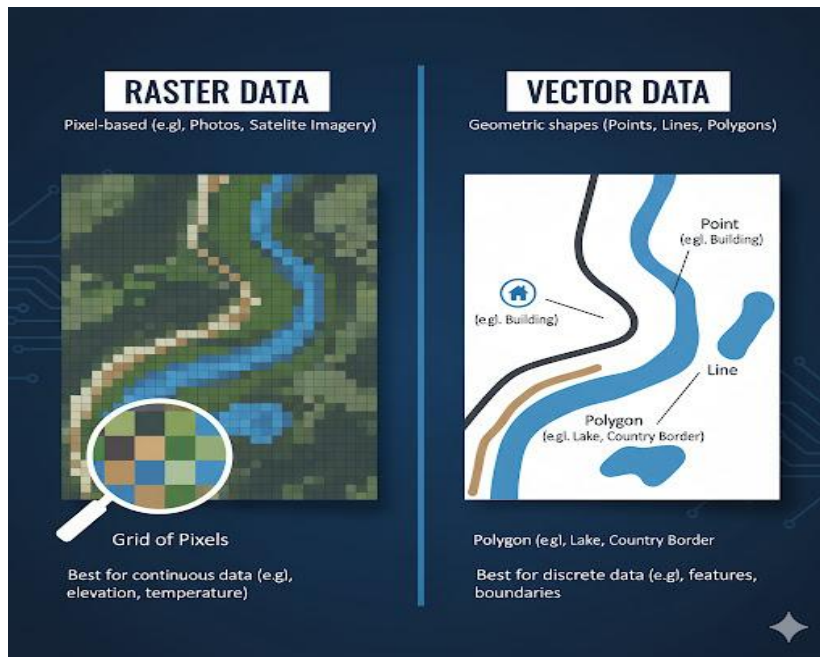
- **Key Idea:** GIS is *more* than just a map. It is a **database that is linked to a map**. It connects *what* (the data) with *where* (the location).
- **Example:** A map is just a picture. A GIS is a *system* where you can *click* on a building (spatial data) and see its *attributes* (non-spatial data: owner's name, number of floors, year built, land value).

11.2 COMPONENTS OF GIS

1. **Hardware:** The computer, monitor, GPS units, scanners, and plotters.
2. **Software:** The program that runs the system (e.g., **ArcGIS** (commercial) or **QGIS** (free/open-source)).
3. **Data:** The "fuel" of the GIS. (See 11.4).
4. **People:** The GIS Analyst—the trained human "brain" who runs the analysis.
5. **Procedures:** The methods and rules for conducting the analysis.

11.3 TYPES OF GEOGRAPHIC DATA IN GIS

This is the most fundamental concept in GIS.



1. Raster Data:

- **What:** A grid of *pixels* (or cells), just like a digital photograph.
- **How:** Each pixel has a *single value*.
- **Best for:** Representing *continuous* data (things that have a value everywhere), such as:
 - Satellite images
 - Elevation (a map of height)
 - Temperature maps
 - Rainfall maps
- **File types:** .jpg, .tif, .img

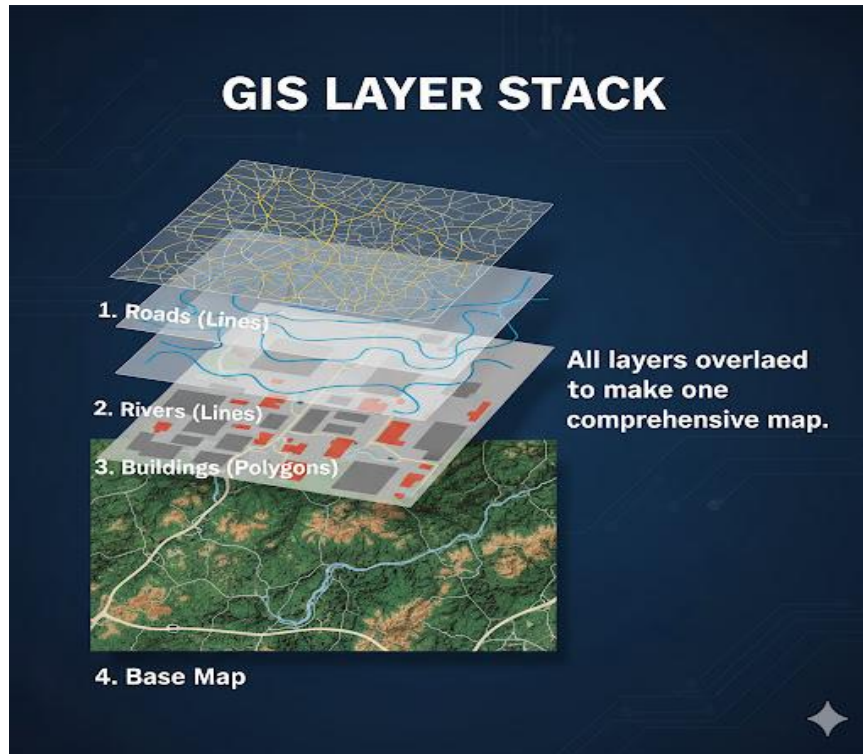
2. Vector Data:

- **What:** A system that represents features using *coordinates* (X, Y).
- **How:** It uses three types of features:
 - **Points:** A single coordinate (e.g., a hospital, a well, a crime location).
 - **Lines (or Arcs):** A series of connected points (e.g., a road, a river, a pipeline).
 - **Polygons:** A closed set of lines (e.g., a building footprint, a Local Government Area (LGA) boundary, a lake, a farm plot).

- **Best for:** Representing *discrete* features with clear boundaries.

11.4 FUNCTIONS OF GIS (SPATIAL ANALYSIS)

GIS is not just for *making* maps; it's for *analyzing* them. This is done by **overlaying layers**.



- **Overlay Analysis:** This is the core function. It involves "stacking" different data layers.
 - *Question:* "Where is the best place to build a new hospital?"
 - *Analysis:* Overlay a layer of **Roads** (for accessibility), a layer of **Population Density** (to be near people), and a layer of **Existing Hospitals** (to find an *underserved* area). The place that meets all three criteria is the answer.
- **Buffering:**
 - *Question:* "Which houses are at risk from this leaking pipeline?"
 - *Analysis:* Create a **buffer** (a polygon) of "500 metres" *around* the pipeline layer. Any house inside this buffer is at risk.
- **Network Analysis:**

- *Question:* "What is the *fastest* route for a fire truck to get from the station to the fire?"
- *Analysis:* The GIS "knows" the road network, speed limits, and one-way streets.

11.5 APPLICATIONS OF GIS (IN NIGERIA)

1. **Urban Planning:** Mapping urban sprawl, planning new roads, zoning land (residential vs. commercial).
2. **Health Services:**
 - **Epidemiology: Mapping disease outbreaks.** (This was famously used by Dr. John Snow to find the source of a cholera outbreak in 1854 London). It is used in Nigeria to track malaria, Lassa fever, and COVID-19 hotspots.
 - **Site Selection:** Finding the best location for a new clinic (see "Overlay Analysis").
3. **Disaster Risk Reduction:** Mapping flood-prone areas (e.g., in Lagos or Lokoja) and planning evacuation routes.
4. **Natural Resource Management:** Mapping deforestation, monitoring oil spills, managing forest reserves.
5. **Security:** Mapping crime "hotspots" to deploy police patrols more efficiently.

11.6 CHALLENGES OF GIS TECHNOLOGY

1. **High Cost:** Commercial software (like ArcGIS) and high-powered computers are expensive.
2. **Lack of Skilled Personnel:** There is a major shortage of trained GIS analysts in Nigeria.
3. **Data Availability and Quality:** This is the *biggest* challenge. "Garbage In, Garbage Out" (GIGO). If the data (e.g., a population map, a road map) is old, inaccurate, or non-existent, the GIS analysis will be useless.

EVALUATION (WEEK 11)

1. Define GIS. What is the key difference between a GIS and a standard digital map?
2. List the five (5) components of a GIS.
3. Differentiate clearly between **Raster** and **Vector** data. What is each best at representing?

4. What is "GIS Layering" (Overlay Analysis)? Give a practical example.
5. What is the *biggest* challenge to the adoption of GIS in Nigeria?

ASSIGNMENT (WEEK 11)

1. You are a GIS analyst for the Lagos State government.
2. You are given the task: "Find the best location to build a new primary school."
3. Write a one-page report describing your "GIS analysis." List at least three (3) *data layers* you would need to overlay to make your decision, and explain *why* each layer is important. (e.g., "Layer 1: Population of children under 10. Layer 2: Location of existing schools...").

